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# Software Requirements Specification

for

## ZipClip

Version 1.0

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# 1 Introduction

## 1.1 Purpose

This Software Requirements Specification (SRS) document defines, describes, and details the complete set of requirements for the proposed system named **ZipClip**. ZipClip is an AI-assisted platform designed to automatically convert long-form videos into short-form clips and story reels optimized for multiple social platforms such as YouTube Shorts, Instagram Reels, and TikTok.

The purpose of this document is to serve as a comprehensive communication bridge between the stakeholders, designers, and developers involved in the project. It outlines the system's intended functionality, performance objectives, design constraints, and user expectations in a clear and structured format. The SRS ensures that all contributors to the development process have a unified understanding of what is to be built and tested.

## 1.2 Document Conventions

- **UI:** refers to the web-based user interface.
- **Clip:** a short video segment extracted from a longer video.
- Items marked as *(optional)* are not required for minimum viable product (MVP).

## 1.3 Product Scope

**ZipClip** is envisioned as a web-based intelligent video editing assistant that automates the transformation of lengthy raw video content into concise, high-impact short clips and story reels tailored for diverse social media platforms. The system combines artificial intelligence, computer vision, and audio processing to identify the most relevant, emotionally engaging, and contextually significant portions of any given video.

The primary goal of ZipClip is to drastically reduce the time, skill, and effort required for manual video editing. Through automated highlight detection and context analysis, the system provides creators with an intuitive interface to preview, refine, and export clips that match platform-specific formats such as portrait ratios, time limits, and resolution requirements.

## 1.4 References

- ZipClip Zeroth Review — internal project documentation.
- SRS\_Career — reference SRS format used for layout alignment.

## 2 Overall Description

### 2.1 Product Perspective

ZipClip is a web application with an AI processing backend. It ingests user video files, analyzes audio/video frames with specialized AI modules, and outputs platform-ready clips and story reels.

### 2.2 Product Functions

Key inputs and outputs:

- **Inputs:** Video files, optional user preferences (target platform, duration), user images for story reels.
- **Outputs:** Ranked short clips, preview interface, downloadable formatted clips (TikTok/IG/YT).

### 2.3 User Classes and Characteristics

- **Content Creators:** Need quick, high-engagement clips.
- **Educators/Trainers:** Want concise lesson highlights.
- **Marketers:** Need short promotional segments preserving key CTAs.
- **Administrators/Developers:** Manage system, models, and deployment.

### 2.4 Operating Environment

- Web app accessible via modern browsers (Chrome, Firefox).
- Backend (AI processing) in Python, deployable on cloud or local servers.
- Storage: object store for videos (e.g., S3 or equivalent), MongoDB or similar for meta-data.

### 2.5 Design and Implementation Constraints

- Must handle large video uploads robustly (chunked upload).
- Processing should be parallelizable; models should be modular.

### 2.6 User Documentation

User guides will include: quick-start for creators, dashboard walkthrough, and an admin guide for model updates and system monitoring.

### 2.7 Assumptions and Dependencies

- Users have internet access and modern browsers.
- Third-party AI frameworks (OpenCV, PyTorch/TensorFlow) available.

## 3 External Interface Requirements

### 3.1 User Interfaces

Responsive web dashboard with:

- Upload panel, clip preview timelines, trimming controls, export options.
- Story reel builder for images + auto-generated transitions and music.

### 3.2 Hardware Interfaces

The user-side hardware requirements are minimal, as all intensive AI processing is handled by the backend servers. The following standard hardware is expected on the client machine:

- **Input Devices:** A standard keyboard and mouse are required for navigating the web interface, managing files, and using the editing controls.
- **Display:** A monitor with a standard resolution (e.g., 1920x1080) is recommended for viewing the dashboard and previewing video content clearly.
- **Network Interface:** A stable internet connection via a network adapter (Ethernet or Wi-Fi) is crucial for uploading large video files and downloading the processed clips in a timely manner.
- **Optional Recording Hardware:** A standard webcam and microphone are necessary only if the user wishes to utilize the optional feature of recording content directly through the browser. No other specialized hardware is required.

### 3.3 Software Interfaces

- Backend API: REST endpoints (Node/Python).
- Database: MongoDB (metadata, user profiles).
- Storage: S3-compatible object store.
- ML frameworks: PyTorch or TensorFlow, OpenCV, MoviePy.

### 3.4 Communications Interfaces

The platform relies on two core communication methods for security and responsiveness.

- **Secure API Communication (HTTPS):** All primary client-server communication, including file uploads and data queries, occurs through a RESTful API. This entire channel is encrypted with HTTPS to ensure all data is transferred securely.
- **Real-Time Progress Updates (WebSockets):** For a dynamic user experience, the system uses WebSockets to push live status updates directly to the user's dashboard. This provides instant feedback on tasks like video processing without requiring the user to refresh the page.

## 4 System Features

### 4.1 Video Ingestion & Preprocessing

- Chunked upload, frame extraction, audio separation, sample-rate normalization. This initial stage prepares the raw video data for subsequent AI analysis.

### 4.2 Multi-Model AI Analysis

- Scene boundary detection, voice activity detection, speech-to-text, emotion recognition, salience scoring. Multiple AI models work in parallel to extract key features from the video and audio streams.

### 4.3 Highlight Extraction

- Rank segments by engagement score; merge adjacent high-score frames to form clips while preserving narrative flow. The system then intelligently selects the most engaging moments to form compelling short clips.

### 4.4 Story Reel Creator

- Convert supplied images into short reels with AI-selected transitions, captions, and soundtrack suggestions. This feature allows users to quickly create dynamic video slideshows from static images.

### 4.5 Platform Optimization

- Output presets for YouTube Shorts, Instagram Reels, TikTok (aspect ratio, duration, bitrate). Final videos are automatically formatted to meet the technical specifications of target social media platforms.

### 4.6 Interactive Dashboard

- Clip preview, manual trimming, reorder, and export; simple UX for non-technical creators. Users have full control to review and refine the AI-generated suggestions before exporting.

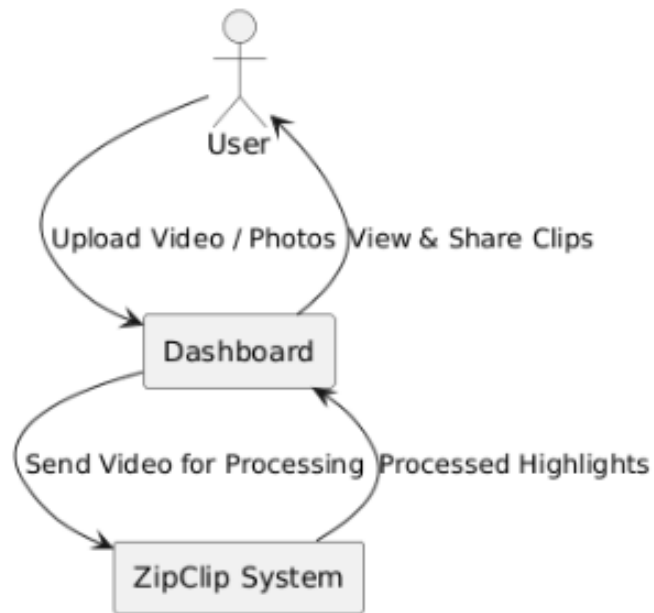


Figure 1: Context Diagram of ZipClip.

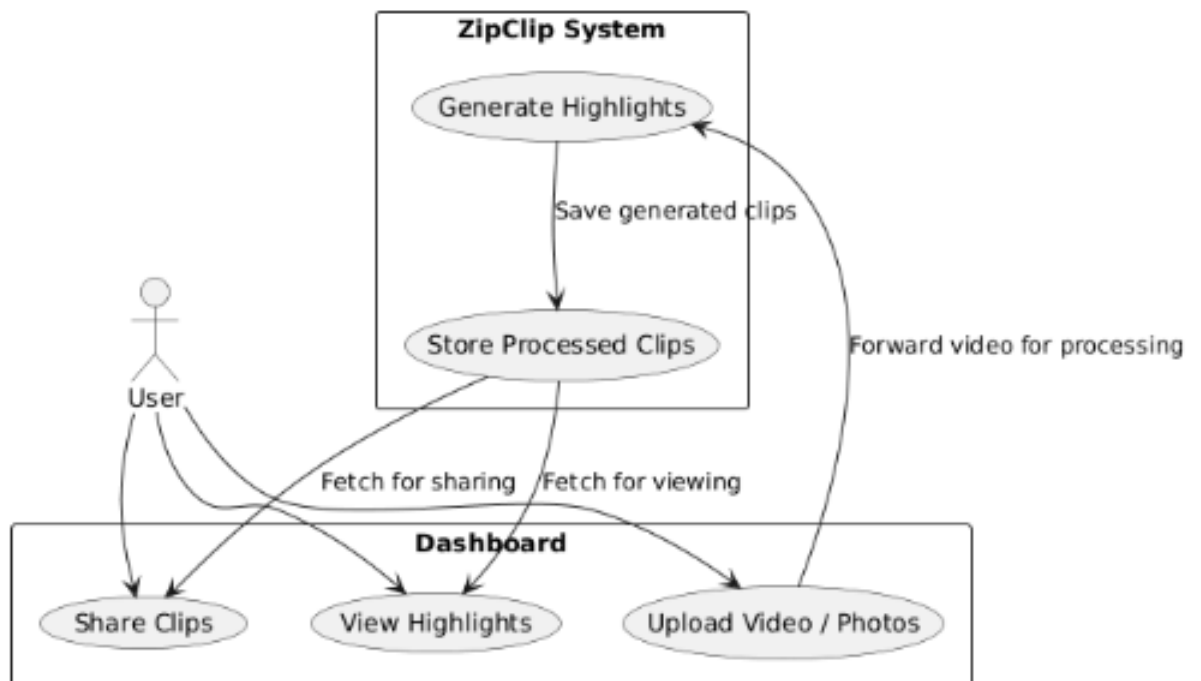


Figure 2: Use Case Diagram of ZipClip.

## 5 Other Nonfunctional Requirements

### 5.1 Performance Requirements

- **Processing Latency:** Video processing is a background task. For a standard 10-minute 1080p video, processing should be completed within a reasonable timeframe, and the user must be notified upon completion.
- **UI Responsiveness:** All user interactions on the dashboard, such as trimming or reordering clips, must be acknowledged by the UI in under 2 seconds. Initial page load times should not exceed 5 seconds on a stable connection.
- **Concurrency:** The system must be designed to handle multiple concurrent video uploads and processing jobs without significant degradation in overall performance.

### 5.2 Safety Requirements

- **Data Integrity:** In the event of a system failure during processing, the user's original uploaded media must not be corrupted or lost. The system should support resuming or restarting failed jobs.
- **Content Advisory:** The application will display clear disclaimers regarding the use of copyrighted material (e.g., music, video clips). Users are responsible for ensuring they have the rights to the content they use and publish.

### 5.3 Security Requirements

- **Authentication:** Access to the system will be controlled via secure user authentication (e.g., email/password with strong hashing or OAuth 2.0).
- **Data Encryption:** All user content will be encrypted both in transit (using TLS 1.2+) and at rest (using AES-256) to prevent unauthorized access.
- **Access Control:** The system must enforce strict authorization rules, ensuring that a user can only access and manage their own uploaded content and data.

### 5.4 Software Quality Attributes

- **Availability:** The service will be designed for high availability, aiming for at least 99.5
- **Usability:** The user interface must be intuitive for non-technical users, requiring a minimal learning curve to go from uploading a video to exporting finished clips.
- **Maintainability:** The backend and frontend will be built using a modular architecture, allowing for easier updates, bug fixes, and replacement of specific AI models without system-wide disruptions.
- **Reliability:** The AI's clip generation must be consistent and dependable, producing relevant and high-quality results for similar types of input videos.



## 6 Other Requirements

### 6.1 Privacy Policies

- **Accessibility and Clarity:** A comprehensive Privacy Policy must be made available and easily accessible from the website footer and user account settings. The policy must be written in clear, simple language to ensure it is understandable to all users.
- **Data Retention and Deletion:** The policy will explicitly define the data retention schedule, specifying how long user-uploaded media and generated clips are stored. It must also detail the process for permanent data erasure upon user request or account deletion.
- **User Control:** Users must be provided with straightforward, self-service tools within their account dashboard to manage their data. This includes options to permanently delete individual files, as well as their entire account and all associated information.

### 6.2 Collection of Personal Information

- **Principle of Minimization:** The system will only collect the absolute minimum personal information necessary for its core functionality. This is limited to a user email address for authentication/communication and an optional display name. No sensitive personal data will be collected.
- **Confidentiality of Media:** All user-uploaded content is treated as confidential. The media is processed by automated systems and is not viewed by human operators unless explicit user consent is given to resolve a technical support ticket.
- **Usage Analytics:** The system may collect anonymized and aggregated usage data for statistical analysis and service improvement. This data will not contain any personally identifiable information and cannot be traced back to an individual user.

### 6.3 Copyrighted Content

- **User Responsibility:** The platform's Terms of Service will clearly state that the user is solely responsible for the content they upload. Users must affirm that they hold the necessary copyrights or have obtained the required licenses for all video and audio they process.
- **Platform as a Tool:** The documentation and terms will specify that ZipClip is a content transformation tool, and the responsibility for avoiding copyright infringement lies with the user as the publisher of the final content.
- **Infringement Claims:** The platform will establish and maintain a clear policy and process for handling copyright infringement claims from third parties, compliant with relevant regulations like the DMCA (Digital Millennium Copyright Act).

## **A   Appendix A: Glossary**

**AI:** Artificial Intelligence.

**NLP:** Natural Language Processing.

**VAD:** Voice Activity Detection.

**ATS:** (Not applicable here) Applicant Tracking System.